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# Palaeomagnetic studies in the Permian Basin of Largentière and implications for the Late Variscan rotations in the French Massif Central

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## SUMMARY

Detailed geological observations and palaeomagnetic analyses were carried out in the Largentière Stephano–Autunian basin and on the Stephanian deposits of the Alès coalfield, both located at the southeastern margin of the French Massif Central. Because of unfavourable rock types, the Alès Stephanian deposits did not yield any results. The palaeomagnetic pole (164.9°E, 45.4°N, $K = 89$, $A_{95} = 4.1°$) deduced from a study of the Autunian sediments of the Largentière Basin agrees very well with the reference pole for stable Europe. The Lodève–Largentière area, that is the southeastern border of the Massif Central, has been stable since Early Permian time with respect to stable Europe, whereas the western part (the Saint-Affrique Rodez Basin and, probably, the Brive Basin) has been rotated counterclockwise.

**Key words:** palaeomagnetism, Permian, regional rotations.

## INTRODUCTION

Previous palaeomagnetic studies in Permian basins of the southwestern border of the French Massif Central (Fig. 1) have indicated the occurrence of counterclockwise rotations in the Saint-Affrique Basin, the Rodez Basin and probably also in the Brive Basin (Cogné *et al.* 1990, 1993; Diego-Orozco & Henry 1993, 1998; Diego-Orozco 1994; Chen *et al.* 1997). The same analysis performed in the Lodève Basin, located at the southeastern edge of the Massif Central, did not show any rotation (Mérabet & Guillaume 1988). To determine the extent of the rotated area, it is therefore important to study other basins situated on the margins of the Massif Central. The Largentière Basin (44.32°N, 4.18°E) is situated at the same latitude as the rotated Rodez Basin, but on the southeastern border. It represents the first important Permian outcrop to the northeast of the non-rotated Lodève Basin. It was chosen as the first objective in a new systematic palaeomagnetic analysis of the Permian basins of the Massif Central because of its key geographical position. It also provides an opportunity to analyse the underlying Upper Carboniferous deposits (Uppermost Westphalian to Late Stephanian) of the Alès coalfield to determine possible rotations between Carboniferous and Permian times. Our study began with a detailed geological analysis because the stratigraphy and structure of the Largentière Basin were insufficiently known (Samama 1969; Gand 1994).

## GEOLOGICAL BACKGROUND

Stephano–Autunian continental siliciclastic deposits are known in the Largentière area, at the northern edge of the Cévennes Massif (Samama 1969; Becq-Giraudon 1993; Gand 1994). The main inlier is a NE–SW-trending graben (Fig. 2). To the north, the deposits are in contact with the basement of Tanargue and southwards they disappear under a thick unconformable cover of sandstone of Triassic age (Gand 1994). They are found at a depth of 417 m in the Morte-Mérie 1 scientific borehole (see Fig. 1), 5 km south of the last Permian surface outcrop (Sureau 1993). Although densely cross-cut by post-Palaeozoic polyphase faults, the inlier has a regional monoclinal structure with an average dip of 20° to the southeast. It is composed of four sedimentary sequences (Fig. 3), the second of which is divided into two informal lithostratigraphic units (Becq-Giraudon 1993).

### Sequence I = Unit I (thickness 0–200 m)

This comprises a stack of fluvio-torrential sequences with very coarse, unsorted conglomerates (mass-flow deposits), conglomerates with rounded, well-sorted, imbricated pebbles, micaceous siltstones and associated coaly streaks. This unit seals a very rough palaeotopography, which explains the recorded variation in thickness (Samama 1969; Fogliérini *et al.*


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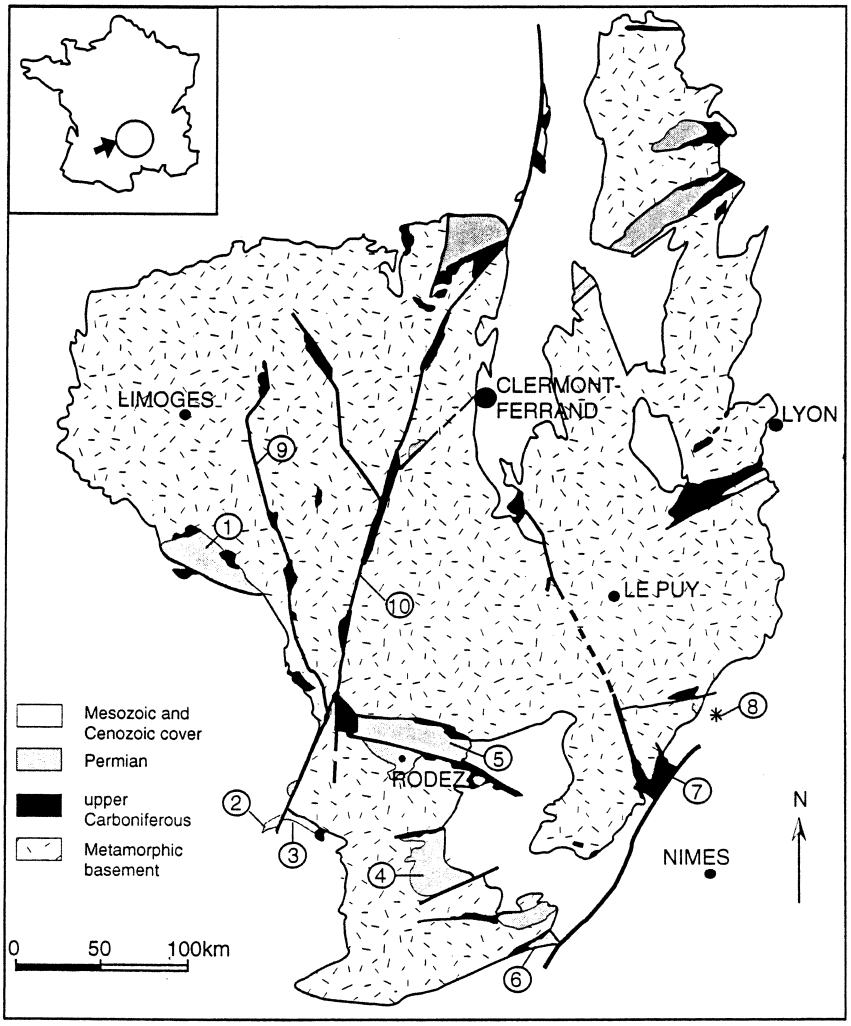



**Figure 1.** Permo-Carboniferous basins of the French Massif Central. 1: Brive Basin; 2: Grésigne dome; 3: Najac/Carmaux inliers; 4: Saint-Affrique Basin; 5: Strait of Rodez; 6: Roujan/Neffiès inlier; 7: Alès Basin; 8: Morte-Mérie 1 borehole; 9: Argentat Fault; 10: Sillon Houiller Fault.

1980). Its age is Upper Stephanian to Lowermost Autunian (Doubinger in Fogliérini *et al.* 1980; Razin & Becq-Giraudon in Sureau 1993; Farjanel 1995).

### Sequence II = Unit II + Unit III

*Unit II (80 m)*

This unit consists of reddish, coarse, sorted conglomerates with erosional bases, coarse- to medium-grained sandstone and micaceous siltstones stacked in fining-upwards sequences. The contact with the underlying Unit I is progressive. The upper limit is conventionally set at the base of the first red bed. Towards the top, siltstone beds become bioturbated by the burrowing Scoyenia ichnosp.

*Unit III (140 m)*

This unit consists of dominant red micaceous siltstones and subordinate, fine- to medium-grained, well-sorted, bedded, erosive sandstone stacked in sequences representing deposits of a distal flood-plain environment. The siltstone beds were intensively bioturbated by Scoyenia ichnosp. which entirely erased any primary sedimentary structure. The passage from Unit II to Unit III is progressive, with a gradual change from dominantly braided river sediments to flood-plain deposits. No palaeontological evidence is available and this sequence is considered to be Autunian merely on lithological evidence.

### Sequence III = Unit IV (100 m)

This is divided into fluvio-lacustrine to palustrine, dominantly grey-coloured sequences (Becq-Giraudon 1993; Razin & Becq-Giraudon in Sureau 1993). The base of this unit is marked by a 10–20 m thick coarse conglomeratic fluviatile sandstone. The top sequence is characterized by a 13 m thick carbonaceous sandstone of palustrine origin (Becq-Giraudon 1993) which has yielded numerous tetrapod tracks and flora which assign an Autunian age to this sequence (Gand 1994).

### Sequence IV = Unit V (60 m)

This comprises sandy and silty fining-upwards sequences, reflecting a distal flood-plain environment. The siltstone beds are bioturbated by vertical, cylindrical tubes but this bioturbation does not erase primary sedimentary structures.

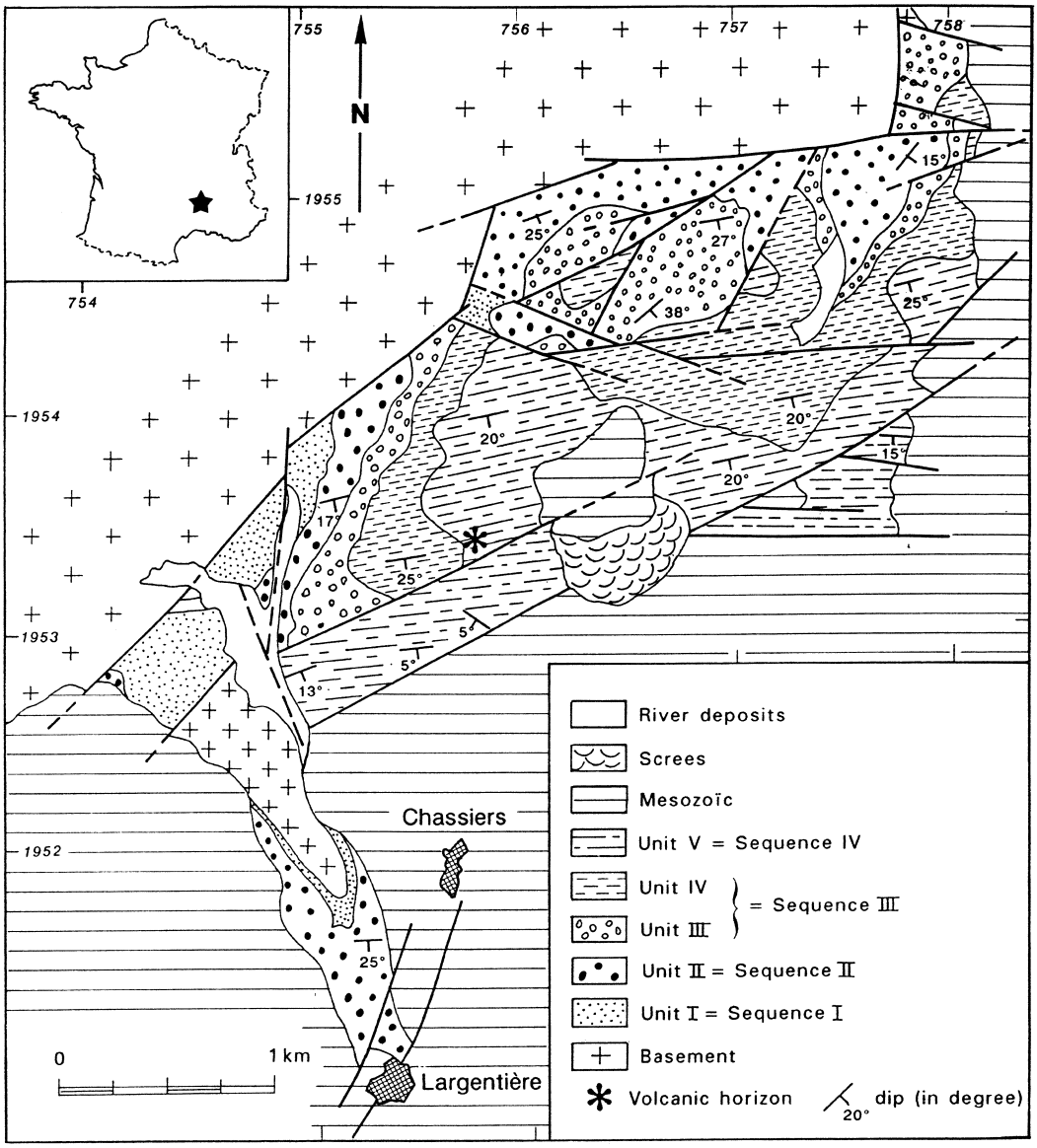



**Figure 2.** Geological map of the Largentière Basin.

The red colour reappears at 4–5 m above the boundary with IV. Locally, at the same level, a 2 m thick volcanic unit occurs. The same tetrapod tracks as those of Unit IV are recorded but are less frequent. Based on this, an Autunian age is given to Unit V.

Based on the results of the Morte-Mérie 1 borehole, the total thickness must have been in excess of 600 m, possibly more than 1000 m as commonly recorded in other Stephano–Autunian basins within the French Massif Central and in other areas (Châteauneuf & Farjanel 1989).

## SAMPLING

109 oriented cores were collected at 13 sites throughout the Largentière graben (Fig. 4). All units were sampled, with the notable exception of Unit I because of the lack of adequate outcrop and unfavourable lithology. 26 samples were also taken from 10 sections (corresponding to sequences III and IV) of the Morte-Mérie 1 oriented drill core (Sureau 1993); the dip of the Permian series at Morte-Mérie 1 is about 10° towards the south. The samples present three main facies, red (only outcrops), grey-brown (only borehole) or grey-black in colour.

## MAGNETIC MINERALOGY

Curie temperature curves for the red facies (Fig. 5), determined using an Agico CS2-KLY2 system, show three different components with Curie temperatures of about 100–120, 580 and 670 °C. The first component gives relatively important variations in the Curie curve. It is barely affected by heating up to 350 °C (small increase of susceptibility), but disappears after 450 °C treatment. It could be goethite (but goethite dehydrates between 250 and 400 °C) or more likely a titanomaghemite with low Curie temperature; this indicates the existence of important volcanic activity in the vicinity of the deposits. This is confirmed by the discovery of a small volcanic unit. The two other components are magnetite and haematite.

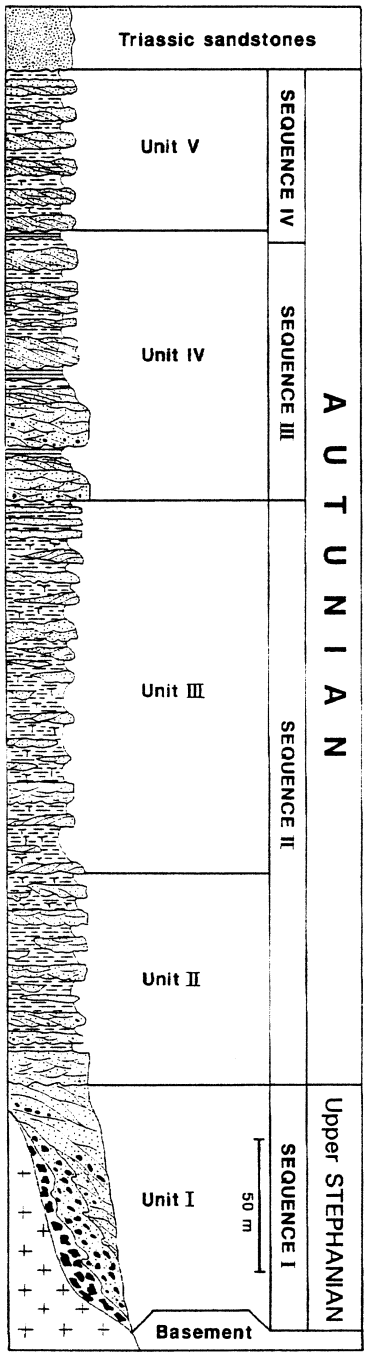



**Figure 3.** Stratigraphic section of the Largentière Basin deposits.

The Curie curve, with partial runs to point out eventual mineralogical alteration, for the grey-brown facies (Fig. 5) shows a moderate decrease up to 100–120 °C; this probably indicates the presence of titanomaghemite, similar to that observed for the red facies. Haematite and magnetite are indicated by the cooling curve. However, mineralogical alteration occurs at low temperatures, and at least a part of the magnetite is the result of this alteration. The fact that the maximum blocking temperature of the remanent magnetization is higher than 580 °C indicates that at least a part of the haematite existed before this alteration.

For the grey-black facies, the Curie curves (Fig. 5) show only mineralogical alteration due to heating. During heating, there is no change up to 320 °C. Partial runs showed that pyrrhotite appears at about 420 °C and magnetite at above 450 °C. Thermal demagnetization of the NRM, however, gives a maximum blocking temperature of about 500–550 °C. Therefore, a very small amount of magnetite must have existed before the thermal treatment.

## PALAEOMAGNETIC ANALYSIS

Prior to any magnetic analysis, all the specimens were put in a zero magnetic field shielded space for more than two weeks in order to reduce the viscous remanent magnetization (VRM) acquired *in situ* and after sampling. The intensity and the direction of the natural remanent magnetization (NRM) were measured on a JR-4 spinner magnetometer (Agico, Brno). Alternating magnetic field (AF) and thermal demagnetization procedures were performed using an AF demagnetizer, which automatically cancels the parasitic anhysteretic remanence (Le Goff 1985) and a magnetically shielded furnace. Both instruments were constructed at the Saint-Maur Laboratory.

The NRM intensity, $J$, is relatively high in the red facies (about $10^{-1}$ A m$^{-1}$) and in the grey-brown facies (from $10^{-1}$ to $10^{-3}$ A m$^{-1}$). It is commonly very low in the grey-black facies (weaker than $10^{-4}$ A m$^{-1}$).

The three facies exhibit different behaviour during demagnetization. However, in all cases a viscous component is first eliminated by thermal treatment up to 250 °C. It is weak in the red facies, but is stronger in the other facies, sometimes being the only component isolated.

The grey-black and, to a lesser degree, the grey-brown facies show important mineralogical alteration during thermal treatment, giving, in particular, a very strong increase of the magnetic viscosity. AF treatment is effective for the samples with the grey-black facies, and was used after heating to 250 °C. It often has little effect on the grey-brown samples. Thus these samples were only demagnetized by thermal treatment. The samples from the red facies were also only thermally demagnetized.

Most of the samples yielded a characteristic remanent magnetization (ChRM) (Fig. 6). However, in several samples of the grey-black facies, the magnetization intensity was too weak to obtain significant data. Some of the grey-brown samples also show important mineralogical alteration during heating. For these two reasons, only four levels from the Morte-Mérie 1 borehole gave significant results. These last data agree well with the outcrop data, confirming the quality of the orienting of these parts of the borehole core. (Two other levels of the Morte-Mérie 1 borehole yield only a low-temperature component—lower than 300 °C—with a vertical direction. This could be an overprint acquired during the drilling.)

All significant results are presented in Table 1 and on Figs 7 and 8.

## ALÈS CARBONIFEROUS BASIN

Situated 40 km southeast of Largentière, the Alès Basin (Fig. 1) is comprised of continental Upper Carboniferous deposits. Recent work shows that its tectonic evolution was controlled by an extensional field stress that rotated counterclockwise during the whole recorded Carboniferous sequence (Djarar

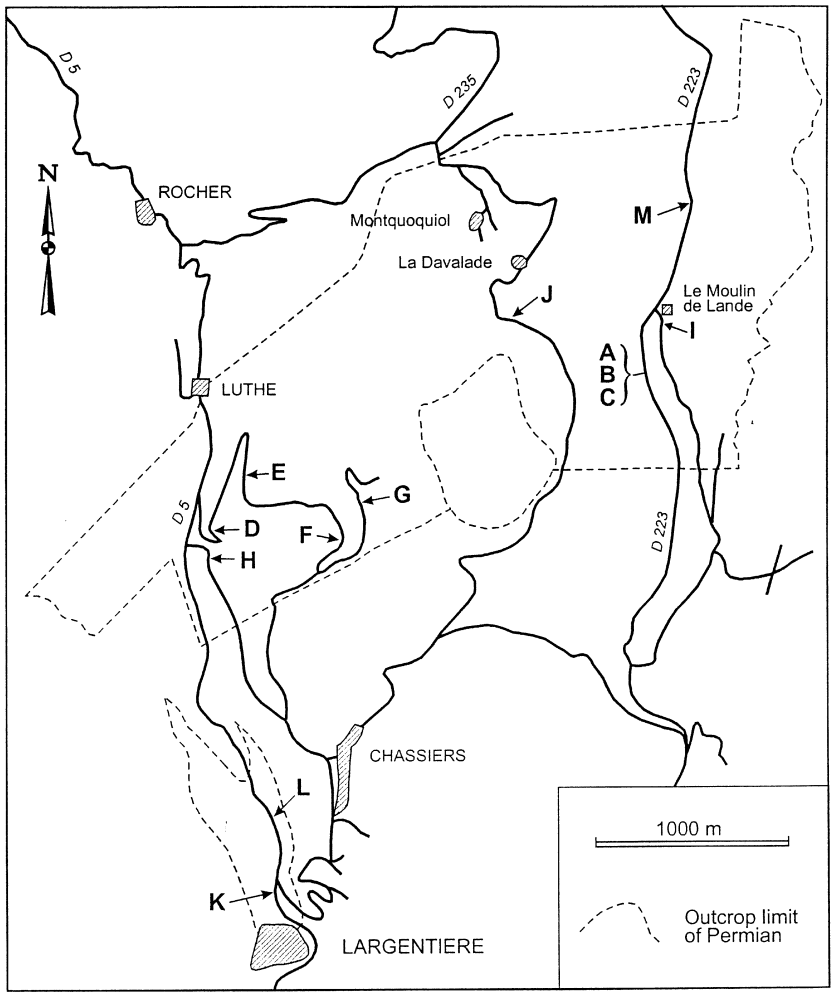



**Figure 4.** Sampling sites (A to M) in the Largentière Basin.

1998). Therefore, it was of fundamental importance to have information on possible rotations in this area before the Autunian, and four sites (two in fine sandstone and two in coaly black shale) were sampled for a preliminary study. The measured NRM has a very weak intensity (lower than $10^{-4}$ A m$^{-1}$) and relatively scattered directions, even within site. During thermal treatment, strong mineralogical alteration appears at temperatures as low as 150–200 °C, and the magnetization becomes unmeasurable because of the very high viscosity and low intensity. AF demagnetization does not yield useful results either, because of the acquisition of ARM before a significant decrease of the magnetization intensity. Thus the study of the Alès Basin was not further developed.

## DISCUSSION

The dip variations are moderate in the Largentière Basin and due mainly to differential tilting of fault blocks (Fig. 2). This tilting is older than Early Triassic, although the age of the unconformable Roubreau sandstone is not precisely constrained (Samama 1969). By reference to the Lodève area, where Upper Permian (Thuringian) red shales and sandstones are unconformably overlain by Lower Triassic sandstone (Saint Martin & Vinchon 1989), the same age range can be assigned to Largentière block tilting. The most convenient precision parameter in this case is $\sqrt{kxky}$ (Henry & Le Goff 1999), $ky$ and $kx$ being, respectively, the maximum and minimum precision parameters associated with the elliptical confidence zone. Its variation during progressive unfolding (Fig. 9) shows that the magnetization was acquired before tilting. The correlation coefficient $r$ between $ky$ and $kx$ values (Henry & Le Goff 1998) points out that the most circular distribution of the mean site directions corresponds to about 100 per cent unfolding. The ChRM was therefore acquired before the Early Triassic and most probably reflects the primary magnetization.

The coincidence between data from the Largentière Basin and the Morte-Mérie 1 borehole shows that no local differential rotation affected the whole area, despite the proximity of the Cévennes faults.

Fig. 10 presents the corresponding palaeomagnetic pole compared to the palaeomagnetic poles from the other Permian basins of the French Massif Central and to the APWP for stable Europe (Van der Voo 1993). This new pole agrees very well with the reference pole for stable Europe. Therefore, no regional rotation relative to stable Europe has affected the Largentière area since Lower Permian times. This is not the case for the two formations from the Saint-Affrique Basin (Cogné *et al.* 1990; Diego-Orozco & Henry 1993; Diego-Orozco

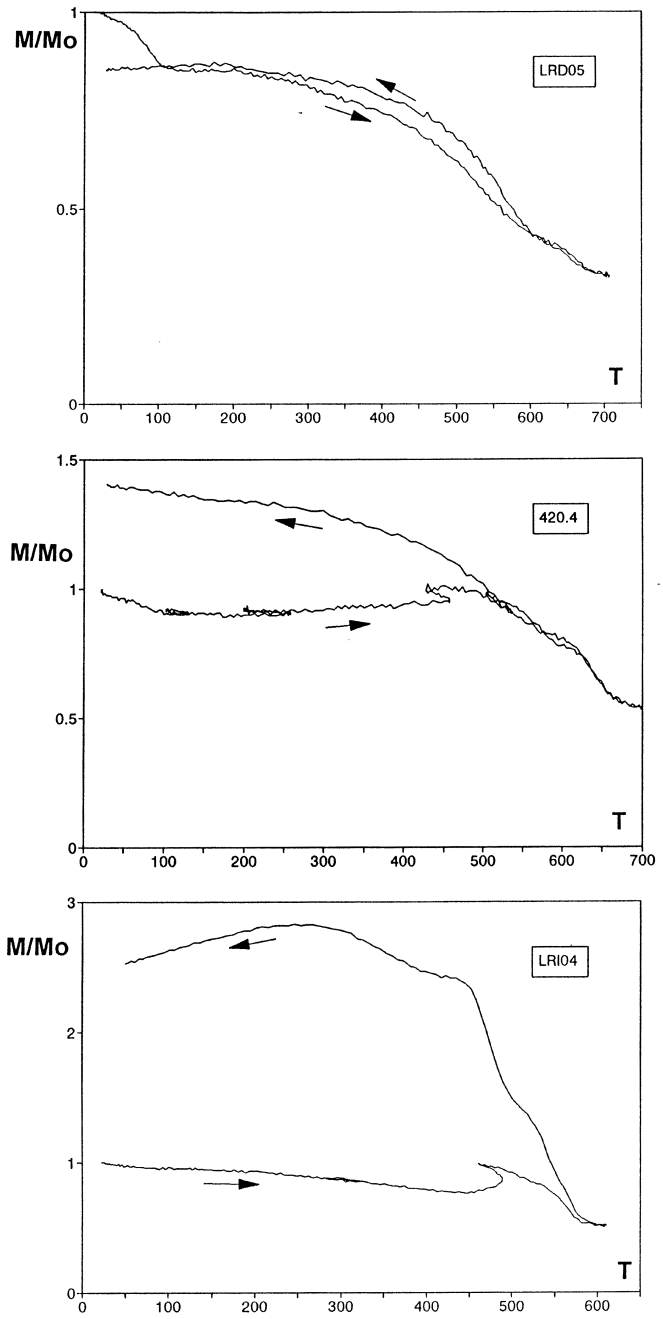



**Figure 5.** Curie curves ($M/Mo$ as a function of $T$ in °C) for the red facies (sample LRD05), the grey-brown facies (sample 420.4) and the grey-black facies (sample LRI04).

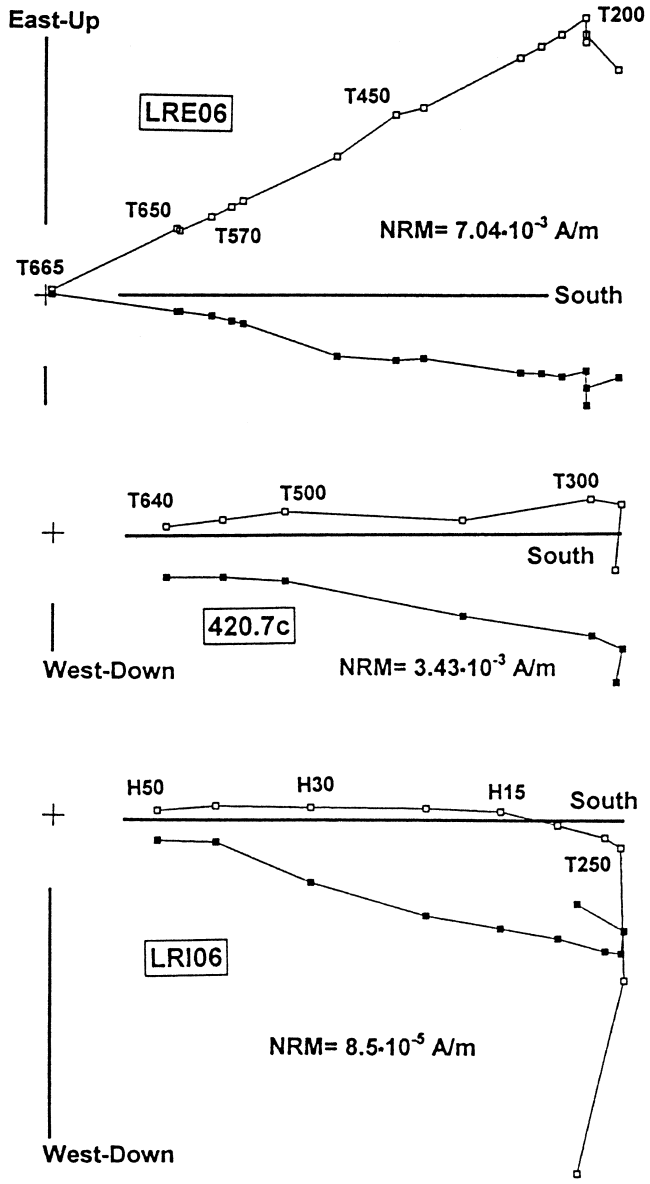



**Figure 6.** Orthogonal vector plot (solid and open symbols correspond, respectively, to horizontal and vertical planes) for the red facies (sample LRE06), the grey-brown facies (sample 420.7c) and the grey-black facies (sample LRI06).

1994), and the 'Grès Rouges' formation of the Rodez Basin (Diego-Orozco & Henry 1998; Diego-Orozco 1994), where a counterclockwise rotation is clear. Taking into account the uncertainties on the APWP (Fig. 10) and the poles (Table 2), the results from the other formations from the Saint-Affrique and Rodez basins and from the Brive Basin (Chen *et al.* 1997) do not yield any conclusions about eventual rotation. However, since all these data are well grouped and the Saint-Affrique and Rodez basins were affected by rotation, the Brive Basin possibly also underwent rotation (see Diego-Orozco & Henry 1998).

Two contrasting hypotheses are proposed in order to explain the late Palaeozoic tectonic evolution of the internal zone of the Variscan belt: the oldest assumes that the Permian basins evolved under an extensional brittle tectonic regime immediately following, and dramatically contrasting with, Upper Carboniferous compressive deformation emphasized by wrench faulting (Arthaud *et al.* 1977; Arthaud & Matte 1979; Bonijoly & Castaing 1987; Genna & Debriette 1996). Recent work has resulted in a quite different hypothesis: in the Variscan belt of the French Massif Central, Devonian–Early Carboniferous crustal thickening (with a final crustal thickness of the order of 60–70 km) resulted in the rapid uplift during the Late Carboniferous of a Tibetan-like high plateau and formation of Himalayan-style mountains, with mean altitude as high as 5000 m or more above sea level (Sabrier *et al.* 1993;

**Table 1.** Palaeomagnetic data from Largentière Basin and Morte-Mérie 1 borehole (names of these sites are the depth in metres in the borehole). Number of sites or samples (sa): N; direction: D, I; associated precision parameters $\alpha_{95}$, $k$; latitude: la; longitude: lo; associated precision parameters ($A_{95}$, $K$) of the palaeomagnetic pole.

| Before dip correction | | | | | | After dip correction | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Site | $N$ | $D$ (°) | $I$ (°) | $k$ | $\alpha$95 (°) | $D$ (°) | $I$ (°) | $k$ | $\alpha$95 (°) | la. °N | lo. °E | $K$ | $A_{95}$ (°) |
| LRA | 5 | 200.0 | 17.9 | 322 | 3.5 | 197.7 | 12.7 | 322 | 3.5 | – | – | – | – |
| LRB | 5 | 200.7 | −0.7 | 171 | 4.8 | 200.2 | 4.6 | 171 | 4.8 | – | – | – | – |
| LRC | 3 | 185.0 | 4.8 | 248 | 5.1 | 184.9 | −3.4 | 248 | 5.1 | – | – | – | – |
| LRD | 10 | 180.2 | −6.3 | 177 | 3.3 | 180.7 | 2.4 | 177 | 3.3 | – | – | – | – |
| LRE | 4 | 189.2 | 3.0 | 109 | 6.7 | 188.8 | −0.2 | 109 | 6.7 | – | – | – | – |
| LRF | 7 | 198.0 | −13.4 | 341 | 2.9 | 198.1 | −9.5 | 341 | 2.9 | – | – | – | – |
| LRG | 6 | 208.8 | 4.4 | 217 | 3.9 | 209.5 | −9.5 | 217 | 3.9 | – | – | – | – |
| LRH | 5 | 182.4 | −28.8 | 244 | 4.0 | 188.0 | −20.2 | 244 | 4.0 | – | – | – | – |
| LRI | 8 | 195.2 | 14.0 | 96 | 5.0 | 194.0 | 0.9 | 96 | 5.0 | – | – | – | – |
| LRJ | 6 | 200.3 | 3.7 | 775 | 2.1 | 200.2 | 1.2 | 775 | 2.1 | – | – | – | – |
| LRK | 7 | 194.7 | 6.1 | 285 | 3.1 | 194.5 | −4.4 | 285 | 3.1 | – | – | – | – |
| LRL | 16 | 191.9 | 5.6 | 278 | 2.1 | 191.6 | −2.5 | 278 | 2.1 | – | – | – | – |
| LRM | 11 | 197.9 | 18.8 | 245 | 2.7 | 198.8 | −5.2 | 245 | 2.7 | – | – | – | – |
| **sites** | **13** | **193.2** | **2.4** | **29** | **7.2** | **193.4** | **−2.7** | **55** | **5.2** | **164.9** | **45.4** | **–** | **–** |
| | **13** | | | | | | | | | **164.9** | **45.4** | **89** | **4.1** |
| **samp** | **93 sa** | **194.0** | **3.4** | **30** | **2.7** | **194.2** | **−2.6** | **54** | **2.0** | **163.8** | **45.2** | **–** | **–** |
| | **93 sa** | | | | | | | | | **163.9** | **45.2** | **84** | **1.6** |
| 420.7 | 3 | 192.4 | 6.8 | 1747 | 1.9 | 189.2 | −1.3 | 1747 | 1.9 | – | – | – | – |
| 441.5 | 4 | 209.8 | 0.5 | 968 | 2.2 | 206.9 | −9.3 | 968 | 2.2 | – | – | – | – |
| 449.8 | 1 | 179.4 | −4.0 | | | 176.6 | −14.0 | – | – | – | – | – | – |
| 462.6 | 1 | 177.2 | 3.7 | | | 174.5 | −7.1 | – | – | – | – | – | – |
| **mean** | **9** | **197.0** | **2.5** | **34** | **8.0** | **194.1** | **−7.1** | **33** | **8.1** | | | | |

**Table 2.** Palaeomagnetic data from the different basins. Rodez [Salabru FIII (1) and 'Grès rouges' (2)—Diego-Orozco (1994); Diego-Orozco & Henry (1998)], Lodève [Upper Autunian (3), Saxonian (4) and Thuringian (5)—Mérabet & Guillaume (1988); Henry (1989)], Saint-Affrique [Saint-Rome 2a from the northern (6) and southern (7) borders of the basin, Saint-Rome 2b (8), Dourdou sandstone (9) and Saint-Pierre pelites (10)—Cogné *et al.* (1990); Diego-Orozco & Henry (1993); Diego-Orozco (1994)] and Brive [Autunian (11) and Saxo-Thuringian (12)—Chen *et al.* (1997)]. For formations 6 and 7, only declination is significant (see Diego-Orozco & Henry 1993) and the data between brackets are only indicative. See Table 1.

| | Before dip correction | | | | | After dip correction | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $N$ | $D$ (°) | $I$ (°) | $k$ | $\alpha$95 (°) | $D$ (°) | $I$ (°) | $k$ | $\alpha$95 (°) | la. °N | lo. °E | $K$ | $A_{95}$ (°) |
| 1 | 6 sites | 193.1 | −19.9 | 16 | 14.0 | 190.9 | −7.1 | 173 | 4.3 | 48.0 | 166.1 | 453 | 2.7 |
| 2 | 19 sites | 188.9 | −17.3 | 33 | 5.6 | 190.1 | −15.6 | 56 | 4.3 | 52.4 | 165.9 | 68 | 3.9 |
| 3 | 5 sites | 192.6 | 17.9 | 442 | 3.0 | 190.6 | 8.1 | 411 | 3.1 | 41.3 | 169.3 | 464 | 2.9 |
| 4 | 3 sites | 190.4 | 15.3 | 351 | 4.3 | 190.1 | 3.5 | 351 | 4.3 | 43.7 | 169.5 | 666 | 3.1 |
| 5 | 6 sites | 198.9 | −0.4 | 1361 | 1.5 | 199.4 | −11.8 | 1361 | 1.5 | 48.6 | 153.5 | 2096 | 1.2 |
| 6 | 6 sites | 196.6 | (11.0 | 124 | 5.1) | 195.9 | (6.9 | 93 | 5.9) | – | – | – | – |
| 7 | 3 sites | 173 | (2.5 | 101 | 8) | 172 | (6 | 37 | 14) | – | – | – | – |
| 8 | 7 sites | 182.8 | −3.9 | 545 | 2.3 | 182.5 | 0.4 | 393 | 2.7 | 45.9 | 179.2 | 978 | 1.7 |
| 9 | 8 sites | 184.4 | 13.1 | 30 | 9.1 | 183.6 | −4.4 | 240 | 3.2 | 48.2 | 177.4 | 590 | 2.0 |
| 10 | 8 sites | 192.3 | −4.3 | 134 | 4.3 | 192.4 | −8.4 | 114 | 4.6 | 48.9 | 163.8 | 325 | 2.7 |
| 11 | 3 sites | 191.2 | −4.9 | 116 | 11.5 | 191.5 | −10.5 | 132 | 11.0 | 48.9 | 165 | 381 | 6.2 |
| 12 | 12 sites | 195.5 | −18.8 | 32 | 7.8 | 192.6 | −10.6 | 86 | 4.7 | 48.8 | 162.2 | 193 | 3.1 |

Becq-Giraudon & Van den Driessche 1994; Becq-Giraudon *et al.* 1996). During the ensuing gravity collapse of the thickened Variscan crust, this plateau evolved into a basin-and-range tectonic province (Menard & Molnar 1988; Van den Driessche & Brun 1989, 1992; Burg *et al.* 1990; Becq-Giraudon & Van den Driessche 1993, 1994; Burg *et al.* 1994; Faure 1995). Two extensional episodes (Namurian–Westphalian and Stephano–Autunian) are identified in conjunction with this thickening–unthickening cycle (Faure & Becq-Giraudon 1993; Becq-Giraudon & Van Den Driessche 1994; Burg *et al.* 1994; Faure 1995), and the Permian basins are a consequence of the second extensional (Stephano–Autunian) episode. A major argument in favour of this hypothesis is the occurrence of a counterclockwise regional rotation, active during most of the Stephano–Autunian sedimentation episode within the Saint-Affrique and Rodez basins and possibly the Brive Basin as well (Cogné *et al.* 1990; Diego-Orozco & Henry 1993, 1998; Diego-Orozco 1994; Chen *et al.* 1997). This rotation was

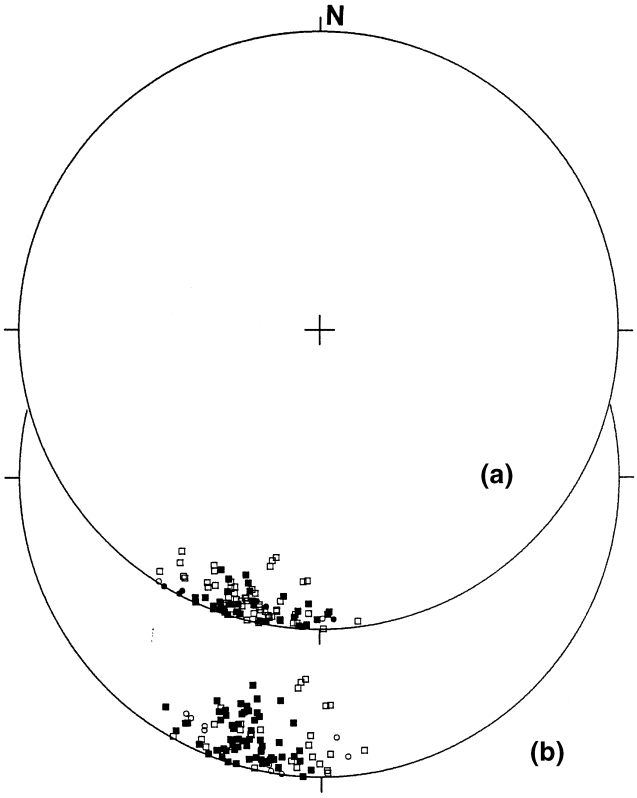



**Figure 7.** ChRM directions before (b) and after (a) dip correction. Data from surface outcrops (squares) and drill cores (circles). Stereographic projection; the solid (open) symbols correspond to the lower (upper) hemisphere.

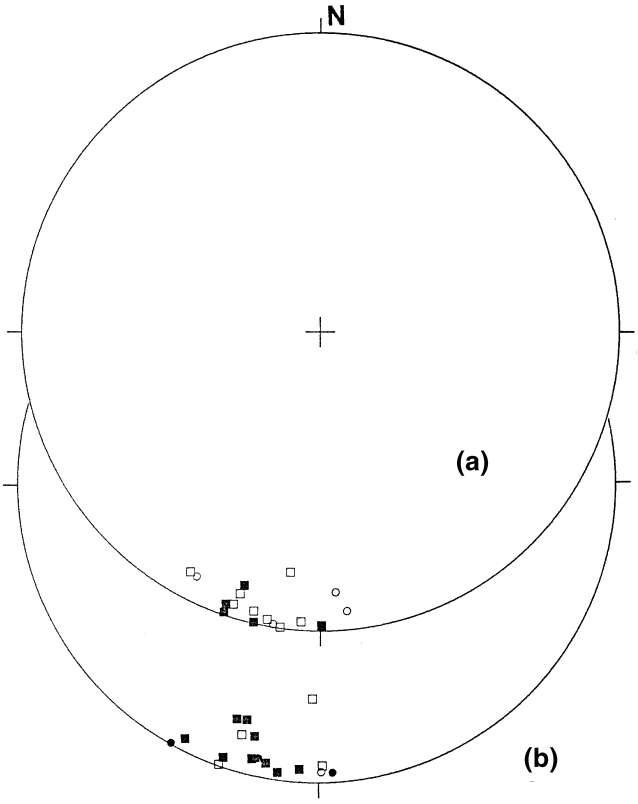



**Figure 8.** Mean site directions before (b) and after (a) dip correction (see Fig. 7).

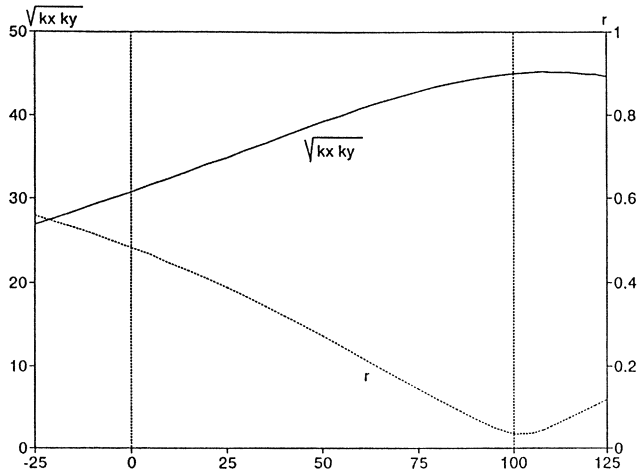



**Figure 9.** Variation of $\sqrt{kxky}$. and $r$ parameters during progressive unfolding (unfolding percentage from −25 to 125).

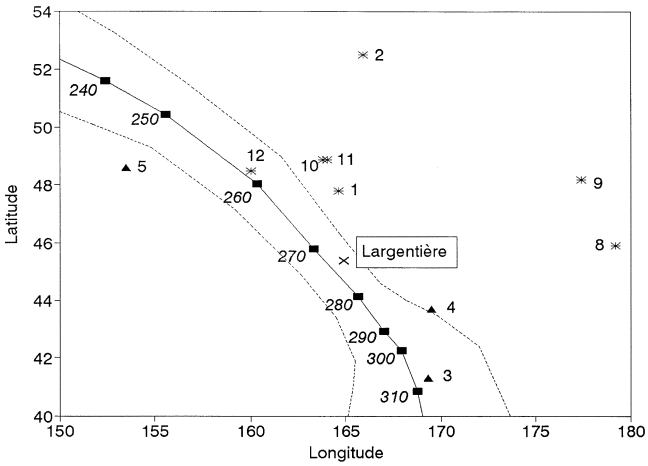



**Figure 10.** APWP for Europe (black squares) with uncertainty and indication of the corresponding age in Ma (number in italics) using the Le Goff *et al.* (1992) method based on a pole selection from Van der Voo (1993), and palaeomagnetic poles from the Lodève (3–5), Saint-Affrique (8–10), Rodez (1–2), Brive (11–12) and Largentière basins. See Table 2.

accommodated by the normal listric border faults of those basins (Legrand *et al.* 1991). In such a geodynamical context, palaeomagnetic data are the only tool available to delineate accurately the extent of the rotated area and the magnitude of rotation. In this case, these data reveal that, contrary to the south-central and southwestern parts of the Massif Central, the southeastern margin did not undergo any regional rotation. This implies that the rotation of the Saint-Affrique–Rodez block must have been accommodated to the west along a fault. The best candidate is the Sillon Houiller sinistral wrench fault, whose role as a transfer fault between adjacent tectonic blocks during the Stephano–Autunian extensional tectonic episode was strongly advocated by Burg *et al.* (1990), Faure (1995) and Chen *et al.* (1997). There are nevertheless still unsolved questions about the eastern limit of the Rodez block (Fig. 11). South of the east–west-trending Alzon Fault (Gèze *et al.* 1985), the rotational movement is easily explained by movement along the border faults of the Saint-Affrique Basin, whereas north of the Alzon Fault, no information is available because

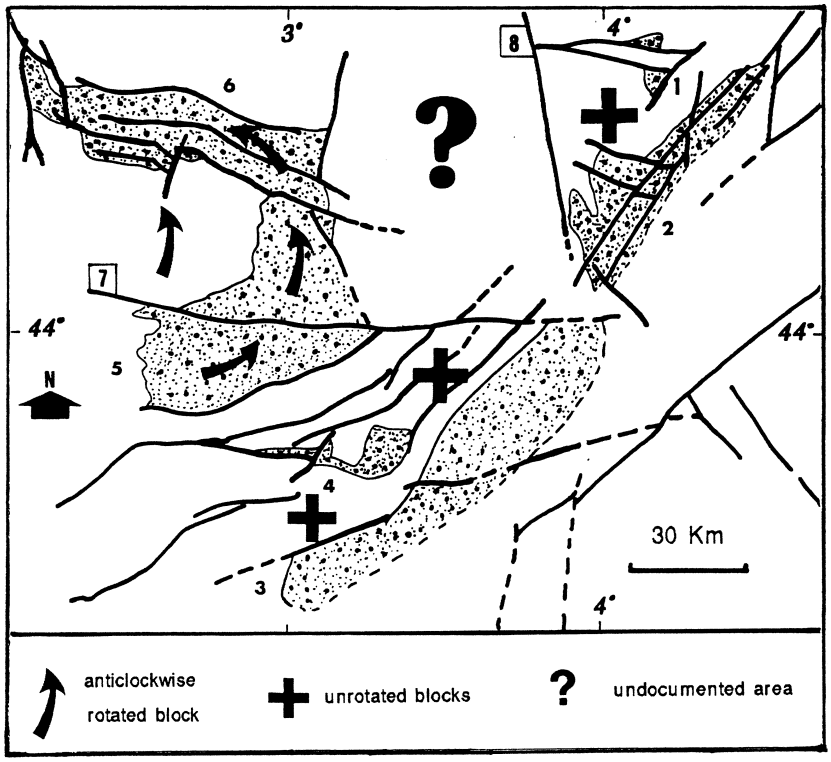



**Figure 11.** Repartition of the known Permo-Carboniferous deposits in the southern French Massif Central (after Debeglia & Gable 1984). 1: Largentière Basin; 2: Alès Basin; 3: Roujan/Neffiès Basin; 4: Lodève Basin; 5: Saint Affrique Basin; 6: Rodez Basin; 7: Alzon Fault; 8: Villefort wrench fault.

of the presence of the thick Mesozoic cover of the Causses, which hides the basement between the Rodez area and the Cévennes Massif. In the Cévennes granito-gneissic massif, the Villefort wrench fault (Fig. 11) has long been considered to be a major crustal fault, responsible for the opening and the tectono-sedimentary evolution of the Upper Carboniferous Alès Basin (Arthaud *et al.* 1977; Bonijoly & Castaing 1987). Recent works (Delénin *et al.* 1988; Delénin 1990; Wang 1991; Djarar 1998) have demonstrated that the Villefort wrench fault is in fact a secondary structure, its main brittle movement (with a maximum horizontal displacement of 12 km) post-Stephanian, and therefore it is unlikely to be capable of accommodating this rotational movement.

The situation in the Brive Basin, which could have rotated in the same counterclockwise direction (Diego-Orozco & Henry 1998), is much more complex because of its position to the west of the Argentat Fault (Feys 1989), itself located to the west of the Sillon Houiller Fault (Fig. 1). Both of these faults have sinistral late horizontal movements. Therefore, with the present state of knowledge, the rotation of the Brive Basin, although having the same direction as the Saint Affrique/Rodez block and being coeval, might be the result of the evolution of a specific block, independent of those forming the inner part of the Massif Central. Therefore, the similarities in evolution would be a convergence rather than a real common history. On the other hand, the increasing eastward rate of Late Palaeozoic extension in the Massif Central (Faure 1995) could also constitute an explanation for the Brive rotation. The overall rotational movement recorded for Saint Affrique and Rodez would be accommodated by both the Sillon Houiller and the Argentat faults (Faure 1995). In this case, Brive would not have a separate history from that of Saint Affrique and Rodez. In this respect, knowledge of the evolution of the Permian Grésigne dome and of the Najac/Carmaux Permian inliers (Bonijoly 1989), each located on a different side of the southern prolongation of the Sillon Houiller Fault (Fig. 1) should be decisive in solving the problem.

## CONCLUSIONS

The new palaeomagnetic pole (164.9°E, 45.4°N, $K = 89$, $A_{95} = 4.1°$) obtained from the Autunian sediments of Largentière Basin is in very good agreement with the Permian reference pole for stable Europe. This shows that the southeastern part of the Massif Central was stable during the Permian, in contrast to the rotated south-central block. The most likely candidate to accommodate the rotational movement of the Saint Affrique and Rodez basins is the Sillon Houiller wrench fault, at least in its southern section. If the southeastern border fault of the Saint-Affrique Basin does account for the counterclockwise movement of this basin, to the north, under the thick Mesozoic cover of the Causses, we do not know the location of the fault which must be present between the rotated Rodez block and the stable Cévennes zone. The proposed rotation of Brive, although similar to that of Saint Affrique and Rodez, could be a consequence of the tectonic evolution of a block distinct from those located further east. However, it seems more likely to be a consequence of the sinistral horizontal displacement of the Sillon Houiller and Argentat faults and therefore is due to the same event that produced the Saint-Affrique/Rodez rotation.

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